RAG (Retrieval-Augmented Generation) Learning Notes

1. What is RAG?

RAG combines retrieval from external knowledge sources with an LLM to generate grounded answers.

Pipeline: Documents → Parsing → Chunking → Embeddings → Vector DB → Retrieval → LLM

2. Loaders

Loaders ingest data from PDFs, websites, databases, CSVs, and text files.

Examples: PyPDFLoader, WebBaseLoader, CSVLoader, DirectoryLoader.

3. Parsing

Parsing extracts clean text, tables, and metadata from raw documents.

Good parsing improves retrieval quality significantly.

4. Chunking

Large documents are split into smaller chunks before embedding.

Common strategies: Recursive, Token-based, Semantic chunking.

5. Embeddings

Embeddings convert text into numerical vectors that capture semantic meaning.

6. Vector Databases

Store and search embeddings efficiently.

Examples: Chroma, FAISS, Qdrant, Pinecone.

7. Retrieval

User query is embedded and matched against stored vectors using similarity search.

8. Generation

Retrieved chunks are added to the prompt and sent to the LLM.

Sample RAG Flow

Documents ↓ Parsing ↓ Chunking ↓ Embeddings ↓ Vector Database ↓ User Query ↓ Similarity Search ↓ Retrieved Chunks ↓ LLM ↓ Answer